

METZ FRESCATY, France

WMO: 070900

Lat: 49.072N

Lon: 6.132E

Elev: 629

StdP: 14.36

Time Zone: 1.00 (EUC)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.3	21.4	10.9	9.8	22.3	15.1	12.1	24.8	26.9	49.3	24.4	47.9	3.4	20	0.517

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	87.7	68.1	84.5	67.0	81.2	65.6	70.2	83.4	68.6	80.8	67.1	78.0	6.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	97.8	74.9	64.5	92.9	73.0	63.0	88.2	71.7	34.5	83.6	33.2	80.9	31.9	78.3	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.6	17.7	15.4	12.9	93.7	5.5	4.8	8.9	97.2	5.7	100.0	2.7	102.7	-1.3	106.1
			12.8	73.1	6.0	1.1	8.5	73.9	4.9	74.5	1.6	75.1	-2.8	75.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	36.2	38.1	44.1	50.4	58.2	63.8	66.8	66.8	59.5	52.1	43.5	37.0
	DBStd	12.88	7.91	7.29	6.71	6.43	6.15	6.22	5.59	5.81	5.28	6.33	6.62	7.85
	HDD50	1717	431	334	205	72	8	0	0	0	1	52	209	405
	HDD65	5240	894	753	649	438	225	98	44	47	179	402	644	867
	CDD50	2244	2	1	20	85	261	413	522	520	286	116	15	3
	CDD65	295	0	0	0	0	14	62	101	103	14	1	0	0
	CDH74	3030	0	0	0	26	197	635	1024	988	155	5	0	0
	CDH80	953	0	0	0	1	33	182	341	368	28	0	0	0
	WSAvg	6.5	7.4	7.3	7.3	6.6	6.3	6.1	6.4	5.6	5.6	5.8	6.3	7.5
	PrecAvg	30.2	2.6	2.2	2.5	2.0	2.8	2.8	2.5	2.6	2.4	2.5	2.7	2.8
(15) Precipitation	PrecMax	41.1	5.4	6.6	6.1	6.3	6.2	5.9	5.0	5.6	5.7	6.9	5.3	6.8
	PrecMin	18.7	1.0	0.5	0.2	0.5	0.9	0.0	0.6	0.4	0.6	0.1	0.4	0.2
	PrecStd	5.5	1.1	1.4	1.2	1.1	1.1	1.4	1.3	1.3	1.3	1.7	1.1	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	58.4	68.6	77.5	84.3	89.7	91.7	97.4	83.9	73.8	61.7	56.0
		MCWB	51.9	49.2	53.3	60.2	67.8	67.7	69.6	68.8	66.1	63.6	55.1	52.1
	2%	DB	52.8	54.4	63.3	72.7	78.9	85.0	87.3	88.0	78.8	68.6	57.8	53.2
		MCWB	49.8	48.2	51.8	57.2	63.8	67.2	67.7	68.5	64.6	60.5	53.6	50.2
	5%	DB	50.1	51.4	58.8	68.1	75.5	81.3	84.0	84.0	74.1	65.2	55.3	50.7
		MCWB	47.2	47.1	50.0	54.9	61.2	66.1	66.7	66.8	62.4	59.0	52.0	48.0
	10%	DB	47.5	48.6	55.2	63.7	71.5	77.2	80.3	79.5	70.3	61.8	52.9	48.1
		MCWB	44.9	45.4	49.1	53.1	59.2	64.1	65.0	65.3	60.4	57.2	50.2	45.8
	0.4%	WB	52.5	51.3	56.0	61.5	68.7	71.8	71.9	71.9	68.2	64.6	56.5	52.8
		MCDB	54.7	55.7	64.1	74.7	81.0	84.3	86.1	87.1	78.4	70.7	59.5	55.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	50.1	49.3	53.3	58.5	65.8	69.3	70.0	69.9	66.1	62.0	54.2	50.4
		MCDB	52.6	53.1	59.7	70.0	76.7	81.3	83.1	83.8	75.6	67.3	57.3	52.6
	5%	WB	47.5	47.6	51.5	56.0	62.8	67.3	68.2	68.3	63.9	59.7	52.4	48.2
		MCDB	49.9	50.8	57.8	65.3	71.5	78.3	80.4	80.5	71.5	64.3	55.1	50.5
	10%	WB	45.0	45.6	49.4	53.9	60.6	65.3	66.4	66.6	62.0	57.5	50.3	45.8
		MCDB	47.2	48.6	54.4	62.2	69.2	75.2	76.7	77.1	68.5	61.1	52.7	48.0
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	12.1	15.9	19.1	19.1	19.6	19.1	19.3	18.0	15.4	10.1	8.3
		MCDBR	11.1	17.1	23.7	27.7	26.7	26.7	26.3	28.1	25.7	20.4	13.2	10.2
		MCWBR	9.4	12.0	14.1	14.2	12.9	11.7	9.9	10.8	12.8	12.7	9.7	8.6
	5% WB	MCDBR	10.8	14.7	19.0	24.7	23.6	23.7	23.9	23.6	21.2	18.3	12.3	10.0
		MCWBR	9.6	11.1	12.1	13.3	12.2	11.5	10.0	10.1	11.3	12.0	9.6	8.9
(40) Clear-Sky Solar Irradiance	taub		0.312	0.338	0.365	0.396	0.405	0.409	0.405	0.401	0.375	0.359	0.330	0.310
	taud		2.434	2.384	2.316	2.252	2.258	2.279	2.285	2.323	2.374	2.409	2.431	2.417
	Ebn at Noon		237	256	267	270	271	270	270	265	261	244	226	220
	Edh at Noon		20	27	34	40	41	41	40	37	32	26	20	18
(44) All-Sky Solar Radiation	RadAvg		272	508	892	1361	1610	1812	1711	1472	1121	654	311	217
	RadStd		42	91	132	176	177	160	169	155	119	55	33	36

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page